

Verification Plan: 4 bit Alu

1.DUT Overview

The DUT is a 4bit ALU with no clk or reset signals, which performs arithmetic and logic operations on 2 input operands. Which operation is performed gets decided based on a selector input.

The RTL implementation is a black box, Verification is based solely on the given design specification

Signal	Direction	Width	Description
A	Input	4bit	Operand A
B	Input	4bit	Operand B
opp	Input	3bit	Operation Selector
carry	Output	1bit	Carry(add/sub)or 0 for logic opps
Y	Output	4bit	ALU result

Opcode (op)	Operation	Description	Example (A=3, B=2)	Y	carry
000	ADD	$Y = A + B$	$3 + 2$	0101	Carry = 0
001	SUB	$Y = A - B$ (two's complement)	$3 - 2$	0001	Carry = 0
010	AND	Bitwise AND	$3 \& 2$	0010	0
011	OR	Bitwise OR	$3 \mid 2$	0011	0
100	XOR	Bitwise XOR	$3 \wedge 2$	0001	0
101	NOT A	Bitwise negation of A	~ 3	1100	0
110	PASS A	Output A	3	0011	0
111	PASS B	Output B	2	0010	0

2.Verification Objectives

- Verify Carry is set correctly for Add/Sub and forced to 0 for rest
- Verify Y output for all 8 operations across all input combination
- All possible opcodes are defined so no verification for faulty opcode needed

3.Features to Verify

Feature	Test
ADD: $Y = A + B$ (4bits)	All A,B combinations
ADD: carry = MSB of 5bit sum	All A,B combinations
SUB: $Y = A - B \text{ mod } 16$	All A,B combinations
SUB: Carry bit	All A,B combinations
AND/OR/XOR/NOT: Correct Y	All A,B combinations
PASS A: $Y = A$	All A,B combinations
PASS B: $Y = B$	All A,B combinations
INVALID OPCODE: $Y = 0$, Carry=0	None - all 8 opcodes possible are mapped

4.Testbench Architecture

1. Simple behavioral reference model implemented in SystemVerilog based on design specifications to predict expected value
2. Nested for loops iteration over opcodes and A, B operands feed both reference model and DUT
3. Outputs get compared directly - Mismatches categorized in:
 - a. Y mismatch,
 - b. Carry mismatch
 - c. Y & Carry mismatch
4. Per-opcode, reports Mismatches per category. If opcodes has mismatches displays all operands and corresponding Y and carry outputs categorized in Passing and Failing

5.Coverage Goals

Since the design is small enough we aim for a 100% coverage:

Input Space: $8 \text{ ops} * 16 \text{ A} * 16 \text{ B} = 2048$ test iterations

6.Tools

- Simulator: QuestaSim / ModelSim
- Language: SystemVerilog
- Waveform viewer: QuestaSim wave viewer (for the ADD waveform deliverable)
- VCD dump: for waveform capture